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(School of Computing Sciences)

COMP-370 – Software Construction and Development

Category: Computing

Complex Computing Problem (CCP)

Course Title	:	Software Construction and Development	Course code	:	COMP - 370
Batch	:	Fall 2002	Credit Hrs	:	2
Instructor	:	Dr. Malik Nabeel Ahmed Awan	Weightage	:	3%
Semester	:	6th	Time	:	
Date	:	April 18th, 2026	CLO-PLO	:	

CCP Title: Development of Security System using Microservices

1. CCP Statement:

he objective of this project is to design and implement a **distributed microservices-based cybersecurity platform** that ingests, processes, ranks, and serves Indicators of Compromise (IOCs) such as IP addresses and domain names.

The platform will integrate external threat intelligence sources including:

- AbuseIPDB
- AlienVault

Students must extract relevant data (IP addresses and domains), process it through a distributed pipeline using **event-driven architecture**, and evaluate IOC severity using an external ranking API.

The system must utilize:

- Apache Kafka for asynchronous communication
- MySQL for persistent storage

The challenge lies in designing scalable, loosely coupled services that handle real-time data ingestion, transformation, and enrichment.

Failure to properly design service orchestration, data consistency, and asynchronous communication can lead to unreliable threat intelligence processing, making this a **complex computing problem involving distributed systems and real-time pipelines**.

2. Range of Problem-Solving Characteristics

Sr. No.	Characteristic	Description	Mapping
WP1	Range of conflicting requirements	Real-time processing vs data consistency	✓
WP2	Depth of analysis required	Requires architectural design and distributed thinking	✓
WP3	Depth of knowledge required	Needs knowledge of APIs, Kafka, DB, microservices	✓
WP4	Familiarity of issues	Modern distributed systems problem	✓
WP5	Level of problem	Beyond standard monolithic applications	✓
WP6	Stakeholder involvement	Multiple system components interacting	✓
WP7	Consequences	Incorrect ranking impacts threat detection	✓
WP8	Interdependence	Multiple services working together	✓
WP9	Requirement identification	Data flow and system boundaries must be defined	✓

3. Project Description

Students will design a Threat Intelligence Processing Platform composed of 6–9 API and at least 3 – 4 microservices communicating asynchronously via Kafka and synchronically via REST API.

System Workflow

1. Data ingestion from APIs
2. Extraction of IPs and domains
3. Streaming via Kafka topics
4. Storage in MySQL
5. IOC enrichment via ranking API
6. Updating severity scores
7. Querying and analytics services

4. Project Description:

In this project, students will design and implement a **Threat Intelligence Platform** that processes data from external sources and converts it into actionable insights.

The system pipeline will include:

1. Fetching threat data from APIs
2. Extracting relevant IOCs (IP addresses and domains)
3. Streaming data using Kafka
4. Processing and validating data
5. Sending data to a ranking API
6. Receiving severity scores
7. Storing enriched data in MySQL
8. Providing query and analytics capabilities

5. Key Implementation Elements:

The system must follow a microservices architecture with event-driven communication.

Core Components

1. Ingestion Service
   - Connects to external APIs
   - Fetches threat intelligence data
2. Extraction Service
   - Parses JSON responses
   - Extracts:
     - IP addresses
     - Domain names
3. Kafka Producer Service
   - Publishes extracted data to Kafka topics
4. Kafka Consumer / Processing Service
   - Consumes data
   - Validates and filters IOCs
5. Ranking Service
   - Sends IOC to external ranking API
   - Receives severity score
6. Database Service
   - Stores IOC data in MySQL

5. Key Tasks:

Research and Analysis

- Study microservices architecture
- Understand Kafka producer-consumer model

Design

- Define system architecture
- Design database schema

Implementation

- Develop microservices
- Integrate Kafka and APIs

Testing

- Validate data pipeline
- Handle failures and retries

Documentation

- Prepare technical report
- Include architecture and flow diagrams

5. Sustainable Development Goals:

SDG	Mapping
Good Health and Wellbeing	✓
Quality Education	✓
Industry, Innovation and Infrastructure	✓
Peace, Justice and Strong Institutions	✓

6. Expected Learning Outcomes

Students will:

- Gain experience in **distributed system design**
- Learn **event-driven architectures**
- Implement **real-time data pipelines**
- Work with **external APIs and streaming systems**
- Develop **production-like backend systems**